UNIVERSITY INSTITUTE OF COMPUTING

PROJECT REPORT
ON

Student Performance Analysis

Program Name: BCA

Data Science Lab Uaing Excel

26CAP-122

SUBMITED BY:

Name : _____ **UID :** _____

Branch : _____ **Section/ Group:** _____

Semester : _____ **D.O.SUBMITION :** _____

SUBMITED TO:

Name : _____ **D.O.SUBMITION :** _____

Desigination: _____

sign: _____

UNIVERSITY INSTITUTE of COMPUTING
Asia's Fastest Growing University

NAAC GRADE A+ ACCREDITED UNIVERSITY

UNIVERSITY INSTITUTE of COMPUTING
Asia's Fastest Growing University

BONAFIDE CERTIFICATE

Certified that this project report "" is the Bonafide work of "Jaideep Singh" under UID "**26BCD10195**" who carried out the project work under my supervision.

.....

SIGNATURE

UIC DEPARTMENT

ASSISTANT PROFESSOR

RAJAT KAPOOR

Data Science Lab Uaing Excel

Student Performance Analysis

:

1. Aim of the Project:

The aim of this project is to analyse students' academic performance using Microsoft Excel. It helps calculate total marks, percentages, grades and pass/fail results. It also helps identify toppers and compare students' performance through charts and data analysis.

2. Performing Task :

Introduction:

Student Result Analysis is an Excel-based project designed to manage and analyse students' examination results. It uses Excel formulas, functions, tables and charts to calculate marks and evaluate academic performance. It makes result management easier, faster and more accurate.

Objectives:

- To calculate students' total marks and percentages.
- To assign grades based on students' performance.
- To identify toppers and students who need improvement.
- To analyse subject-wise performance.
- To represent results using charts and graphs.

Tools and Technologies Used:

Microsoft Excel

Excel Formulas and Functions

IF() Function

SUM() Function

AVERAGE() Function

COUNTIF() Function

Conditional Formatting

Charts and Graphs

Pivot Tables

-

3. STEPS INCLUDE :

S1) Create a student dataset containing roll numbers, names and subject-wise marks.

S2) Apply Excel formulas to calculate total marks, percentage, grades and pass/fail status.

S3) Analyse the results using sorting, filtering, charts and Pivot Tables to identify toppers and overall performance.

4. Algorithm / Logic / Flow chart :

Algorithm:

Start the project.

Enter student details and subject-wise marks.

Calculate total marks using the SUM() function.

Calculate percentage using the percentage formula.

Assign grades using the IF() function.

Check whether the student has passed or failed.

Identify the highest scorer and calculate average marks.

Create charts to analyse student performance.

Display the final result.

Stop.

Flowchart:

Start → Enter Student Data → Calculate Total Marks → Calculate Percentage → Assign Grade → Check Pass/Fail → Analyse Results → Create Charts → Display Output → End.

5. Code Overview

Microsoft Excel formulas and functions are used instead of traditional programming code.

1. Calculate Total Marks:

=SUM(C2:G2)

2. Calculate Percentage:

=H2/500*100

3. Calculate Grade:

=IF(I2>=90,"A+",IF(I2>=80,"A",IF(I2>=70,"B",IF(I2>=60,"C",IF(I2>=40,"D","F"))))

4. Pass/Fail Status:

=IF(I2>=40,"Pass","Fail")

5. Calculate Average Marks:

=AVERAGE(C2:G2)

6. Find Highest Marks:

=MAX(H2:H21)

These formulas help automate calculations and make student result analysis easier and more efficient.

6. OUTPUT :(SCREENSHOT)

	A	B
1	STUDENT PERFORMANCE ANALYSIS	
2	Project objective	Analyse student marks, calculate totals and percentages, assign grades, identify pass/fail status and visualise performance.
3	Workbook contents	Student Data: sample records and automatic calculations. Subject Analysis: subject-level statistics. Dashboard: key results and charts.
4	How to use	Replace the sample names and marks in the blue input columns on Student Data. Formula columns calculate automatically in Excel.
5	Subjects	English, Mathematics, Computer Science, Accountancy and Economics. Each subject is marked out of 100.
6	Pass rule	A student must score at least 33 in every subject to pass.
7	Grade rule	A+ = 90% and above; A = 80–89.99%; B = 70–79.99%; C = 60–69.99%; D = 40–59.99%; F = below 40%.
8	Important	The dataset is fictional and is provided for academic demonstration. Open the workbook in Microsoft Excel or a compatible spreadsheet app to recalculate formulas.
9		
10	Jaideep Singh	
11	26BCD10195	
12	3- B	

	A	B	C	D	E	F	G	H	I	J	K
1	Roll No.	Student Name	English	Mathematics	Computer Science	Accountancy	Economics	Total Marks	Percentage	Grade	Result
2	101	Aarav Sharma	52	68	86	68	82	356	71.2	B	Pass
3	102	Simran Kaur	60	72	95	55	92	374	74.8	B	Pass
4	103	Rahul Verma	72	93	74	53	74	366	73.2	B	Pass
5	104	Ananya Singh	75	57	92	42	42	308	61.6	C	Pass
6	105	Arjun Mehta	65	79	69	46	51	310	62	C	Pass
7	106	Priya Gupta	92	90	56	20	86	344	68.8	C	Fail
8	107	Kabir Malhotra	89	91	98	68	81	427	85.4	A	Pass
9	108	Mehakpreet Kaur	70	44	63	76	73	326	65.2	C	Pass
10	109	Ishita Patel	34	49	83	65	43	274	54.8	D	Pass
11	110	Dev Kumar	49	93	43	71	52	308	61.6	C	Pass
12	111	Sana Khan	77	88	63	67	73	368	73.6	B	Pass
13	112	Harnoor Singh	51	53	84	44	72	304	60.8	C	Pass
14	113	Rohan Das	97	23	57	46	91	314	62.8	C	Fail
15	114	Aditi Joshi	85	93	84	90	90	442	88.4	A	Pass
16	115	Manav Bansal	79	88	46	49	77	339	67.8	C	Pass
17	116	Jasleen Kaur	72	77	94	66	54	363	72.6	B	Pass
18	117	Vivaan Shah	63	51	98	76	68	356	71.2	B	Pass
19	118	Navya Rao	60	65	70	35	66	296	59.2	D	Pass
20	119	Gurpreet Singh	66	59	43	95	47	310	62	C	Pass
21	120	Diya Kapoor	52	87	87	73	42	341	68.2	C	Pass
22	121	Karan Gill	73	46	77	80	89	365	73	B	Pass
23	122	Tanya Sethi	75	56	78	80	28	317	63.4	C	Fail
24	123	Aditya Nair	49	53	84	52	82	320	64	C	Pass
25	124	Pooja Yadav	64	95	63	64	90	376	75.2	B	Pass
26	125	Reyansh Jain	80	61	57	48	52	298	59.6	D	Pass
27	126	Manya Arora	46	94	53	91	69	353	70.6	B	Pass
28	127	Sahil Chawla	65	52	85	44	85	331	66.2	C	Pass
29	128	Khushi Sharma	75	84	57	68	52	336	67.2	C	Pass
30	129	Yuvraj Sandhu	89	79	42	98	64	372	74.4	B	Pass
31	130	Avni Mishra	73	74	66	84	57	354	70.8	B	Pass

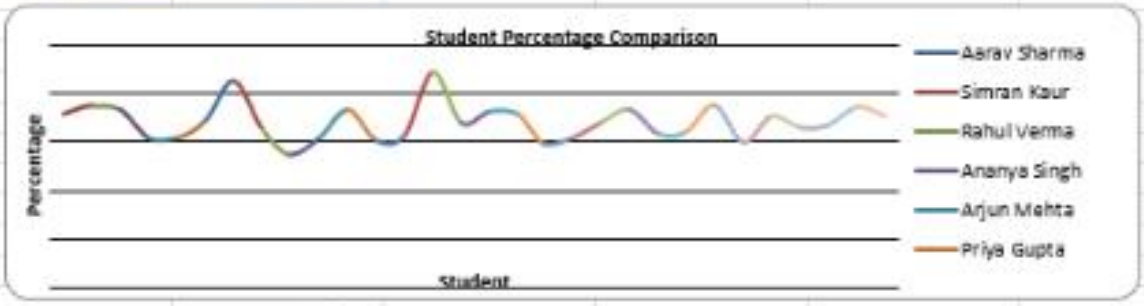
SUBJECT-WISE PERFORMANCE ANALYSIS

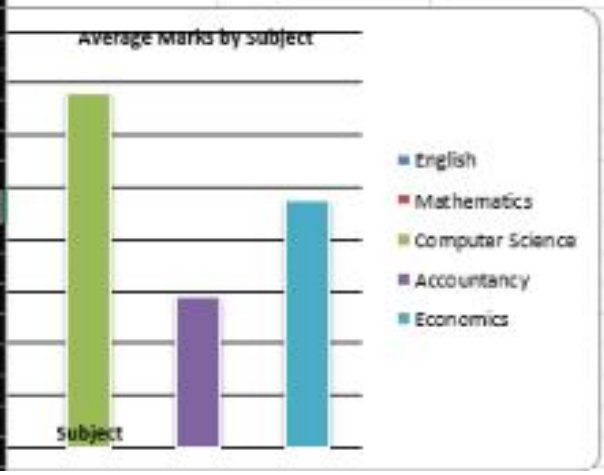
Subject	Average Marks	Highest Marks	Lowest Marks	Students Passed	Pass Rate
English	68.30	97	34	30	100.0%
Mathematics	70.47	95	23	29	96.7%
Computer Science	71.57	98	42	30	100.0%
Accountancy	63.80	98	20	29	96.7%
Economics	67.47	92	28	29	96.7%

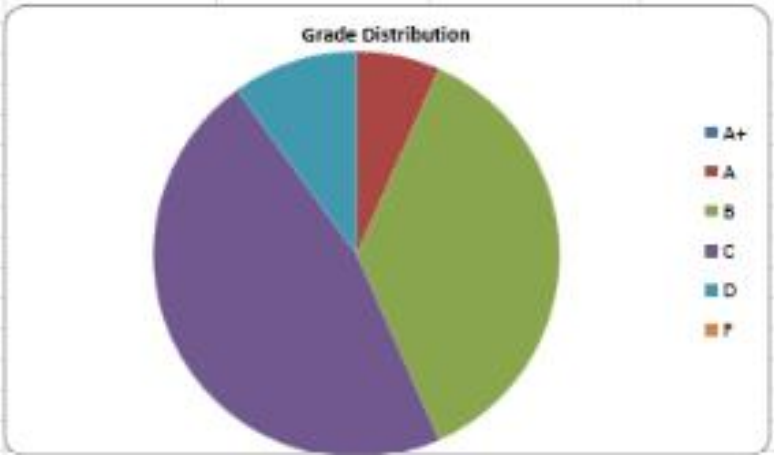
STUDENT RESULT ANALYSIS DASHBOARD

30	Class Average	68.32%	Percentage	88.40%	Percentage	90.0%
Aditi Joshi	Topper Percent	88.40%				

Number of Stud
0
2
11
14
3
0







7. Conclusion:

This project successfully demonstrates how Microsoft Excel can be used for student result analysis. By using formulas, functions, charts and Pivot Tables, student results can be calculated and analyzed easily. It saves time, reduces manual errors and provides a clear understanding of academic performance.

8. LEARNING OUTCOMES:

1. Learned how to organise and manage student data in Excel.
2. Understood the use of Excel formulas and functions.
3. Developed skills in calculating percentages and assigning grades.
4. Learned to create charts and dashboards for data visualisation.
5. Improved analytical and problem-solving skills through practical data analysis

10 . Screenshots

11. Evaluation Grid :

Sr. No.	Parameters	Marks Obtained	Maximum Marks
1.	PROJECT TITLE		2 Marks
2.	DESIGN & IMPLEMENTATION		5 Marks
3.	Github Link		1 Marks
4.	Linkedin Blog Link		1 Marks
5.	Portfolio link		1 Marks
	TOTAL		10 Marks
	AVG		6 Marks

Teacher Signature